AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM354
Software Engineering

Midterm Examination (MTA) – Makeup
Spring Semester 2024/2025

Date: 22 /Apr./2025

Number of Exam Pages: 3 (including this cover sheet)	**Time Allowed:** 2 Hours

Instructions:

- Total Marks: **60**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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Part I – Multiple Choice Questions — 6 Marks

*Answer **all** the following multiple-choice questions. Each question is worth 2 Marks.*

1. A module which provides a service may require other services that are called:

a. abstract dependencies	b. mutual dependencies	c. context dependencies	d. coupled dependencies

2. To which category and type do you classify the requirement:
"The form must accept the mobile number with the country code"

a. Functional, Business	b. Functional, Technical	c. Non-functional, Operational	d. Non-Functional, Security

3. Which UML diagram describes the objects of the system and the relationships between them:

a. Activity diagram	b. Use Case diagram	c. Class diagram	d. Object diagram

Part II – Short Answer Questions — 24 Marks

*Answer **all** the following short answer questions. Each question is worth 6 Marks.*

1. In iterative requirements gathering, one approach to prioritisation is the MoSCoW (Must, Should, Could, Won't) scheme. Discuss the importance of such a scheme and how it could be applied.

2. Discuss malleability as a characteristic that affects software development and whether malleability might increase the possibility of errors or reduce them.

3. For each of the following requirement, identify if it is a functional or non-functional one and state its type.
   a. Learning the basic operation of the system should not take more than 30 minutes for any university student.
   b. The system should use an OTP as part of a 2-Factor authentication when logging in from a new device.
   c. The system must provide the client with the most recent edition of the publication.
   d. The system should provide seniors with a discount in accordance to the country's law.

4. While modelling a system for a hospital, the software engineer came up with the class doctor that has attributes: firstName, lastName, specialisation. Explain why the developer might add more attributes during implementation.

Part III – Problem Solving Questions — 30 Marks

*Answer **all** the following problem-solving questions. Each question is worth 15 Marks.*

1. Develop a Use Case diagram for a Restaurant system with the following requirements:

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a. The main actor is a waiter that can perform the tasks: place order, confirm order, cancel order, settle order bill.
b. A manager is a waiter that can perform all the above tasks in addition to: apply discount.
c. Using pre-condition and post-condition of the use case descriptions, show how "apply discount" cannot be performed unless "confirm order" was performed first.

Note: Make sure to use Generalization if possible.

2. Develop an activity diagram for the task of "Booking a Flight". Make sure to cover the following requirements:
   - Use Swimlanes to represent 2 roles, the Passenger and the FlightSystem
   - Use Synchronization Bars (Fork and Join) at least once
   - Use Decision and Merge nodes at least once
   - Make sure to use at least 6 activities

End of questions

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